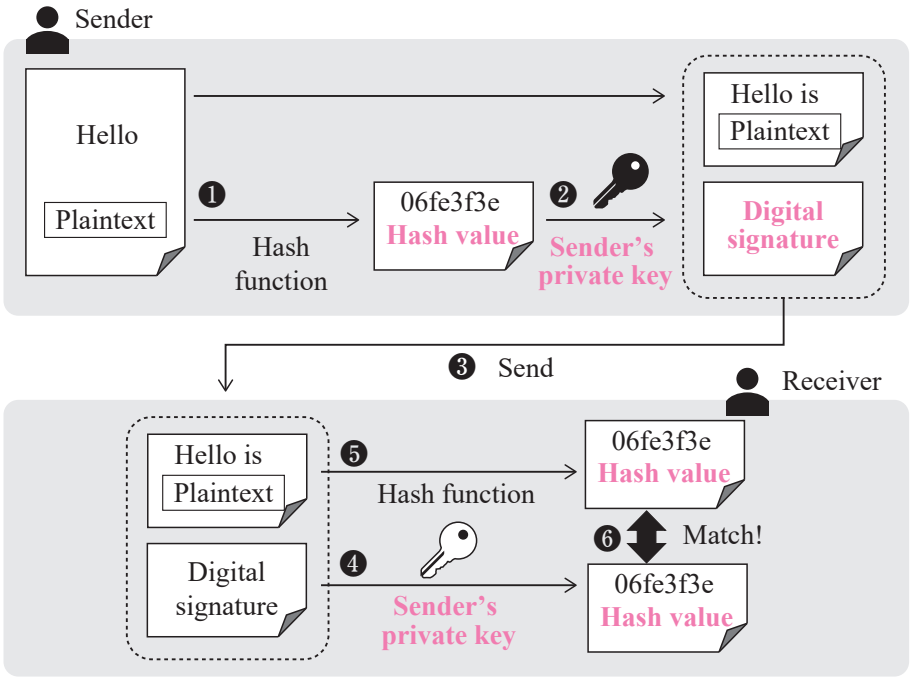
3-5 Information Technology for Safety [2]

Point!

1 Digital Signature

(1) ([1]**Hash function**) : A function that calculates a unique value based on the input data.The value generated by the hash function is called a ([2]**hash value**). It is not possible to restore the original data from the hash value.

(2) ([3]**Digital signature (Electronic signature)**) : A technology that uses public key encryption and hash values to prove that the transmitted data is from the sender and has not been tampered with.



【Sender's Procedure】

1. Use a hash function to generate a ([4]**hash value**) from the plaintext to be sent.
2. Encrypt the hash value using the ([5]**sender's private key**). This encrypted hash is called a ([6]**digital signature**).
3. Send both the plaintext and the digital signature to the recipient.

【Recipient's Procedure】

4. Use the ([7]**sender's public key**) to decrypt the received digital signature and retrieve the original hash value.
5. Use the same hash function used in step 1 to generate a new ([8]**hash value**) from the received plaintext.
6. Compare the two ([9]**hash values**) from steps 4 and 5. If they match, it proves that the message is from the sender and has not been tampered with.

(3) ([10]**Certification authority (CA)**) : A trusted third-party organization that verifies whether a public key truly belongs to its claimed owner.
It issues ([11]**digital certificates**) that include the public key and identifying information of the key owner.

2 SSL / TLS

([12]**SSL/TLS**) : A technology used to encrypt communication between a web server and a web browser. The ([13]**session key method**) is used in this encryption. The URL of an encrypted web page begins with "([14]**https**):// …".

＊TLS (Transport Layer Security) was introduced as a more secure version of the originally used SSL (Secure Sockets Layer). However, since the term "SSL" became widely recognized, the combined term SSL/TLS is often used.

Warm Up

Answer the following questions.

(1) For the following statements **A** to **D** about hash values, mark "○" if the statement is correct, or "×" if it is incorrect.

A If one character in the original data is changed, only one character in the resulting hash value will change.

B A digital signature is created by encrypting the hash value of the document to be sent with a private key.

C It is difficult to restore the original message from its hash value.

D Even if the documents being sent are different, the hash values obtained from the same hash function will always be the same.

(2) The following sentences describe the steps of creating a digital signature. Fill in the blanks [1] to [4] with the appropriate terms.

> The sender generates a ([1]) based on the data they want to send and encrypts it using their ([2]). This is called a ([3]), and the sender sends both the data and the digital signature to the recipient. The recipient decrypts the received ([3]) using the sender's ([4]) to recover the original ([1]). The recipient also generates a ([1]) from the received data using the same hash function. If the two ([1]) values match, it proves that the data is from the sender and has not been tampered with.

(3) What is the name of the technology used to encrypt communication between a web server and a web browser?

Explanation

(1) **A**. If even a single character in the original message is different, the resulting hash value becomes completely different. Therefore: ×

B ○

C ○

D If the message being sent is different, the resulting hash value will also be completely different even when using the same hash function. Therefore: ×

(2) [1] hash value [2] private key [3] digital signature (electronic signature) [4] public key

(3) SSL/TLS

Try

Answer the following question.

(1) Choose the terms that best fit into the blanks [1] to [6] from the options **A** to **J** below, and answer using the letters.

> To allow the recipient to verify that the data was created by the actual sender and was not tampered with during transmission, there is a technology called ([1]). A ([1]) is created by generating a ([2]) from the plaintext to be sent using a program, and then encrypting it with the ([3]). This is attached to the plaintext and sent to the recipient. The recipient decrypts the ([1]) using the ([4]). If the resulting ([2]) matches the ([2]) generated from the received plaintext, it proves that the data was created by the sender and has not been altered. However, this alone cannot prevent impersonation. Therefore, a third-party organization called a ([5]) issues a ([6]) to guarantee that the public key truly belongs to the sender.

A Sender's private key **B** Sender's public key **C** Sender's shared key
D Recipient's private key **E** Recipient's public key **F** Recipient's shared key
G Digital certificate **H** Digital signature **I** Hash value
J Certification Authority (CA)

(2) Choose the terms that best fit into the blanks [1] to [2] from the options **A** to **D** below, and answer using the letters.

> Regarding the hash function used in digital signatures: The same data is always converted into the same ([1]) hash value, and it is ([2]) impossible to restore the original data from the converted hash value.

A Different **B** Same **C** Possible **D** Impossible

(3) Choose the terms that best fit into the blanks [1] to [6] from the options **A** to **G** below, and answer using the letters.

> When the beginning of a web page URL is "https://", it means that encryption using ([1]) is being performed. In ([1]), encryption is carried out using a ([4]) that combines ([2]) and ([3]). In addition, ([1]) also helps prevent phishing to direct towards a fake website by attaching a ([6]) issued by a ([5]).

A Public key encryption **B** Symmetric key encryption **C** Session key method
D Digital certificate **E** Digital signature **F** SSL/TLS
G Certification Authority (CA)

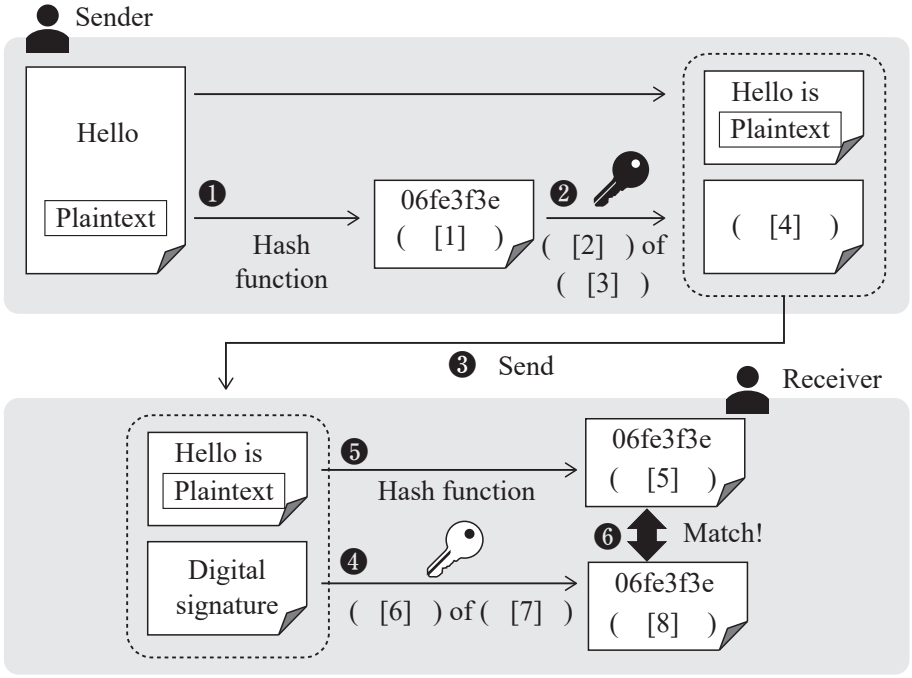
(4) Among the following options **A** to **D**, choose the one that correctly describes a function of SSL/TLS. Answer using the corresponding letter.

A Generates one-time passwords for user authentication on websites.
B Encrypts communication between the web browser and the web server.
C Filters communication to unauthorized websites.
D Detects viruses that spread through networks.

Exercise

1 Answer the following question.

(1) The diagram below illustrates the mechanism of a digital signature. For blanks [1] to [8], choose the appropriate terms from the options **A** to **H** below and answer using the corresponding letters. The same letter may be used more than once.



【Word Bank】 **A** Sender **B** Recipient **C** Shared key **D** Private key
E Public key **F** Hash value **G** Digital signature **H** Electronic authentication

(2) Among the options **A** to **D** below, choose the one that appropriately describes something related to an email with a digital signature. Answer using the corresponding letter.

A Garbled text is more likely to occur during transmission of the email.
B It allows you to confirm whether the email was sent from the correct sender.
C It prevents the contents of the email from being intercepted during transmission.
D It prevents the contents of the email from being lost.

(3) Among the options **A** to **D** below, choose *all* that correctly describe functions of SSL/TLS. Answer using the corresponding letters.

A SSL is the predecessor of TLS, and currently, TLS is the mainstream standard in use.
B It is a function that restricts access to harmful or illegal websites based on certain conditions.
C URLs that begin with “http://...” are encrypted using SSL/TLS.
D SSL/TLS encrypts communication using a session key method that combines symmetric and public key encryption methods.